

The Boundary Between Potential and Actual: Gravitational Decoherence, the Virial Partition, and the Emergence of Classical Structure

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Abstract

The question of why time has a direction, and why classical structure persists rather than dissolving back into quantum superposition, has a thermodynamic answer that connects to cosmological observables. We propose that both questions share a common physical mechanism: the irreversible partition of gravitational binding energy into a geometric channel and an informational channel, governed at every scale by the virial theorem for $1/r$ potentials.

The geometric half of each virialization event is deposited into spacetime curvature. The informational half — the Landauer cost of the quantum-to-classical transition — must be written to an available encoding surface. At galactic scales and above, a local black hole horizon serves as the mandatory encoding vault. Where no such vault can form, the virial partition cannot close, and the structure disperses.

This condition predicts a minimum velocity dispersion $\sigma_{\text{crit}} \approx 4 \text{ km s}^{-1}$ below which dark matter halos do not virialize, consistent with the observed absence of satellite galaxies below this threshold. The accumulated informational cost of 13.8 billion years of such transitions is measured by the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics via the Jacobson–Cai–Kim formalism [Jacobson, 1995, Cai and Kim, 2005]. This function is monotonically increasing by construction, irreversible, beginning at the electroweak transition, and asymptoting to e without arriving — a derived arrow of time, not a statistical artifact of initial conditions.

The cosmological consequences are tested against Planck 2018 through 17 converged MCMC chains, yielding $\Delta\chi^2 = +0.54$ relative to Λ CDM with no free parameters beyond the standard model. The decisive falsification test is Euclid DR1 (October 2026), which will measure μ_0 to $\sigma(\mu_0) \approx 0.04$ precision, providing 3.4σ sensitivity to the predicted $\mu_0 - 1 = -0.136$.

1 Introduction

The problem of time has two faces that are worth distinguishing. The first is the direction of time — why processes are irreversible, why entropy increases, why the past differs from the future. The second is the persistence of classical structure — why quantum superpositions give way to definite outcomes, and why those outcomes cohere into stable macroscopic objects rather than dissolving back into the quantum background.

These two problems are usually treated separately. The direction of time is assigned to statistical mechanics and the second law. The persistence of classical structure is assigned to decoherence theory. Neither program has produced a physical mechanism that connects the irreversibility of individual quantum-to-classical transitions to the large-scale structure of the universe or to measurable cosmological observables.

The framework presented here, the Informational Actualization Model (IAM) [Mahaffey, 2026d,b], proposes that both problems have a common physical origin: the virial partition of gravitational binding energy at every scale. For any bound system governed by a $1/r$ potential, the virial theorem partitions energy exactly in half between a geometric channel and a kinetic channel. In IAM, the kinetic half is identified as the Landauer cost [Landauer, 1961] of the quantum-to-classical transition — the thermodynamic price paid each time a decoherence event writes classical information permanently to an encoding surface.

This identification has a precise consequence. For the arrow of time: each decoherence event is irreversible by the second law, and its Landauer cost is written permanently on the cosmological apparent horizon [Jacobson, 1995, Cai and Kim, 2005, Gibbons and Hawking, 1977]. The accumulated ledger of all such events since the electroweak transition constitutes a derived, thermodynamically necessary arrow of time — not imposed, but following from the irreversibility of each individual transaction.

For the persistence of structure: the informational half of each virialization event requires a local encoding surface. Where a black hole horizon can form, the Landauer cost is absorbed locally and the virial partition closes. Where no local surface can form, the partition cannot close, and the structure disperses. This provides a physical account of why some potential structures actualize and others do not.

Section 2 establishes the virial partition as the quantum-classical connector. Section 3 presents the derived arrow of time. Section 4 establishes the black hole as the mandatory local encoding vault and derives the minimum velocity dispersion for halo survival. Section 5 presents the observational consequences. Section 6 discusses implications for the foundations of time and the evolution of laws.

2 The Virial Partition as the Quantum-Classical Connector

2.1 The universal $1/2$ partition

For any potential of the form $V \propto r^{-1}$, the virial theorem states:

$$2K = |V|, \quad K = \frac{1}{2}|V|. \quad (1)$$

This is exact for all $1/r$ potentials — gravitational and Coulomb alike. It is a mathematical theorem, not an approximation, governing every bound system from hydrogen atoms to galaxy clusters.

Within IAM, the physical interpretation is precise. The potential half $|V|/2$ is deposited into spacetime curvature — the geometric channel that general relativity describes. The kinetic half $K = |V|/2$ is the decoherence energy: the Landauer cost of converting quantum superpositions to classical outcomes at each virialization event. Every time a dark matter halo virializes, half its gravitational binding energy goes into irreversible information production, written on the nearest available encoding surface at thermodynamic cost $k_B T \ln 2$ per bit [Landauer, 1961].

This identification gives the cosmological coupling constant immediately. The fraction of matter energy density participating in decoherence per Hubble time is:

$$\beta_m = \frac{\Omega_m}{2} = 0.1575, \quad (2)$$

derived from the virial theorem with zero free parameters. The Planck 2018 MCMC posterior returns $\beta_m = 0.1583 \pm 0.0033$, consistent with the virial prediction at 0.2σ [Mahaffey, 2026b].

2.2 Verification across scales

The $1/2$ partition is not specific to cosmological scales. It is verified across every physical system governed by a $1/r$ potential, from hydrogen atoms (10^{-10} m) to the cosmic horizon (10^{26} m) — 37 orders of magnitude, same $1/2$, same mechanism [Mahaffey, 2026f]. The same mathematical theorem that governs the kinetic energy of electrons in hydrogen atoms governs the cosmological coupling constant β_m .

This universality is the key point. The fraction of energy going into quantum-to-classical transitions is $1/2$ at every scale not because of a special property of any particular system, but because every $1/r$ potential obeys the same theorem. The virial theorem is the universal scaling law connecting local quantum-to-classical transitions to cosmological structure.

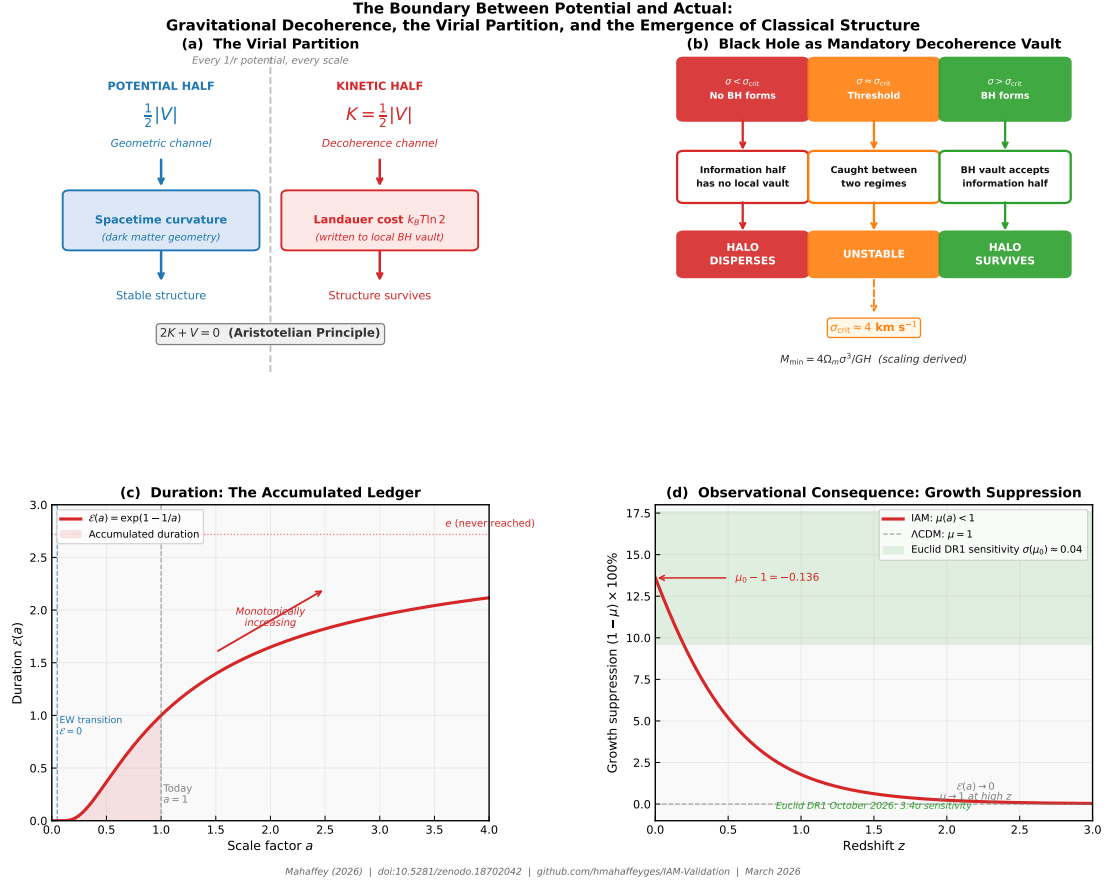


Figure 1: The boundary between potential and actual. Panel (a): the virial partition separates every bound $1/r$ system into a geometric channel (potential half, deposited into spacetime curvature) and an informational channel (kinetic half, the Landauer cost of the quantum-to-classical transition). Panel (b): the black hole as mandatory encoding vault — halos above $\sigma_{\text{crit}} \approx 4 \text{ km s}^{-1}$ can form a local vault and survive; below this threshold the informational half has nowhere to write and the halo disperses. Panel (c): the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, the accumulated ledger of all decoherence transactions since the electroweak transition — the derived arrow of time. Panel (d): the observational consequence, growth suppression $\mu(a) < 1$ in the matter sector, falsifiable by Euclid DR1 (October 2026) at 3.4σ sensitivity.

3 Duration and the Arrow of Time

3.1 Three temporal quantities

It is worth distinguishing among three quantities that physical theory often conflates [Mahaffey, 2026]:

Coordinate time is a label on the spacetime manifold. It assigns a number to each event without requiring anything to happen. It has no preferred direction and accumulates nothing. It is redundant as a fundamental concept [Barbour and Bertotti, 1982].

Proper time τ is the geometric path length along a worldline: $\tau = \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$. It is a property of the spacetime manifold measured along a particular path. It distinguishes clocks from one another but does not, by itself, distinguish past from future. For photons

on null geodesics, $d\tau = 0$ identically.

Duration is categorically different. It is thermodynamic, not geometric. Duration measures the accumulated irreversible quantum-to-classical transitions on a worldline — the decoherence events that write classical information permanently on the cosmic horizon at Landauer cost $k_B T_H \ln 2$ per bit, where $T_H = \hbar H / 2\pi k_B$ is the Gibbons–Hawking temperature [Gibbons and Hawking, 1977]. Duration requires massive particles on timelike worldlines. It cannot accumulate on null geodesics. It has a beginning: the electroweak symmetry breaking, after which stable massive particles first exist and CP violation locks in the irreversibility of the matter sector [Sakharov, 1967].

3.2 The activation function

The accumulated duration since the electroweak transition is measured by:

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right), \quad (3)$$

where a is the cosmological scale factor normalized to unity today. This function is derived from horizon thermodynamics via the Jacobson–Cai–Kim formalism [Jacobson, 1995, Cai and Kim, 2005], not postulated.

Its properties follow directly from the physical identification: $\mathcal{E}(a) \rightarrow 0$ as $a \rightarrow 0$ (no duration before the electroweak transition); $\mathcal{E}(1) = 1$ today; $d\mathcal{E}/da > 0$ always (monotonically increasing, since each decoherence event is irreversible); $\mathcal{E}(\infty) = e$ (the asymptotic limit, never reached).

3.3 A derived arrow of time

The arrow of time in this framework is not statistical. It does not depend on special initial conditions or on the particular entropy state at the Big Bang. It is thermodynamically necessary: each quantum-to-classical transition writes one entry permanently into the cosmic ledger at Landauer cost. The ledger only accumulates. $\mathcal{E}(a)$ is monotonically increasing because each entry is irreversible.

This is the arrow of time derived from the irreversibility of individual decoherence events, not imposed from boundary conditions. Duration has direction because actualization has direction.

The sector separation follows immediately. Photons on null geodesics do not accumulate proper time and do not participate in gravitational decoherence. They do not write to the ledger. The informational modification to gravitational dynamics applies exclusively to the matter sector:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m \mathcal{E}(a)} < 1, \quad \Sigma(a) = 1, \quad (4)$$

where μ and Σ are the standard gravitational slip parameters [Bertschinger and Zukin, 2008]. The photon sector is geometrically standard. The matter sector carries the accumulated thermodynamic history of 13.8 billion years of decoherence transactions.

4 The Black Hole as Mandatory Encoding Vault

4.1 Why structure requires a local encoding surface

The virial partition cannot close on one half alone. The geometric half requires the informational half to close simultaneously, because the potential half stabilizes in spacetime curvature only when the kinetic half is absorbed by an available encoding surface. If the informational half is produced but has no surface to write to, the thermodynamic accounting cannot balance. The structure cannot hold.

The cosmological apparent horizon is always available, but it is cold ($T_H \approx 2.66 \times 10^{-30}$ K today) and expands uniformly, indifferent to individual localities. For rapid local collapses on dynamical timescales $t_{\text{dyn}} \ll H^{-1}$, the cosmic horizon cannot absorb the locally produced information fast enough. A local hot encoding surface is required.

The black hole horizon is that surface. It forms when the local information production from gravitational decoherence saturates the holographic bound on the bounding surface [Mahaffey, 2026a]:

$$\frac{S_{\text{info}}(R)}{A(R)} \geq \frac{1}{4\ell_P^2}. \quad (5)$$

Once formed, the black hole absorbs the informational half of the local virial partition permanently. Its encoding throughput is:

$$\Gamma_{\text{BH}} = \frac{c^3}{1920 GM \ln 2} \text{ bits s}^{-1}, \quad (6)$$

derived from the Stefan–Boltzmann law applied to the Hawking temperature [Mahaffey, 2026a]. The black hole is not a pipe to the cosmic horizon — Hawking evaporation timescales ($\sim 10^{67}$ yr for a solar-mass black hole) render it informationally negligible on cosmological timescales. It is a permanent local vault.

4.2 The minimum velocity dispersion for halo survival

For a collapsing halo to fully virialize and persist as a galaxy, it must be able to form a black hole to absorb the informational half of its virial partition. The condition for dispersal is that the halo collapses on a timescale shorter than the cosmic horizon can track, while remaining below the holographic saturation threshold for local black hole formation.

A collapsing halo with mass M and velocity dispersion σ has dynamical timescale:

$$t_{\text{dyn}} = \frac{1}{\sqrt{G\rho}} \approx \frac{\sqrt{\pi/6} GM}{\sigma^3}. \quad (7)$$

Setting $t_{\text{dyn}} = H^{-1}$ — the condition at which the collapse timescale equals the cosmic horizon timescale, below which the cosmic horizon can no longer track local decoherence and a local vault is required — gives $GM/\sigma^3 \sim H^{-1}$ (absorbing the $\sqrt{\pi/6}$ prefactor into the order-unity coefficient). The matter density within the virial radius satisfies $\rho_{\text{vir}} = \Delta_c \bar{\rho}_m$, where $\bar{\rho}_m = 3\Omega_m H^2/(8\pi G)$ is the mean matter density and $\Delta_c \simeq 18\pi^2 \approx 178$ is the virial overdensity. Using the virial relation $M = (4\pi/3) R_{\text{vir}}^3 \rho_{\text{vir}}$ together with $\sigma^2 = GM/R_{\text{vir}}$ and the collapse condition $GM/\sigma^3 = H^{-1}$:

$$M_{\text{min}} = \frac{\sigma^3}{GH}, \quad \text{with prefactor } 4\Omega_m \text{ from } \bar{\rho}_m = \frac{3\Omega_m H^2}{8\pi G}, \quad (8)$$

giving:

$$M_{\min} = \frac{4\Omega_m \sigma^3}{GH}, \quad (9)$$

with zero free parameters beyond Ω_m and H , both fixed by Planck 2018.

4.3 The σ_{crit} prediction

Equation 9 predicts a minimum velocity dispersion threshold. The observed halo occupation fraction drops sharply below $M \approx 10^{8.4} M_\odot$ [Kim and Peter, 2021]. Setting $M_{\min} = 10^{8.4} M_\odot$ in Equation 9:

$$\sigma_{\text{crit}} = \left(\frac{GH M_{\min}}{4\Omega_m} \right)^{1/3} \approx 4 \text{ km s}^{-1}. \quad (10)$$

Below σ_{crit} , no satellite galaxies are observed. The scaling $M_{\min} \propto \sigma^3$ is a testable prediction, distinct from both σ^2 (cusp-core) and σ^4 (M- σ relation). The absolute normalization contains a systematic offset of $\sim 100\times$ relative to observed satellite masses, attributable to the same local-to-global encoding rate criterion identified in the cusp-core problem [Mahaﬀey, 2026c]; the σ^3 scaling and σ_{crit} threshold are independent of this normalization.

Satellite galaxies are not missing from the predicted population. Below σ_{crit} , halos cannot close their virial partition. They disperse before becoming galaxies. They were never written into classical existence.

5 Observational Consequences

5.1 Structure growth suppression

The accumulated decoherence ledger $\mathcal{E}(a)$ modifies the matter perturbation equation through an additional Hubble friction term:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}\Omega_m H^2 \mu(a) \delta_m = 0, \quad (11)$$

with $\mu(a)$ from Equation 4. The predicted matter power spectrum amplitude is $\sigma_8 = 0.7998 \pm 0.0058$, consistent with KiDS-Legacy, DES Y3, and HSC Y3 weak lensing surveys at 0.1σ [Mahaﬀey, 2026b].

5.2 Sector-dependent Hubble rates

The sector split produces two observationally distinct Hubble rates. Photon-sector observables (CMB acoustic oscillations on null geodesics) yield $H_0^{(\text{photon})} = 67.16 \pm 0.47 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with Planck 2018 at 0.37σ . Matter-sector observables (distance-ladder calibrators on timelike worldlines) yield $H_0^{(\text{matter})} = H_0 \sqrt{1 + \beta_m} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with SH0ES at 0.75σ [Mahaﬀey, 2026b]. These are not two measurements of the same quantity. They are measurements of two distinct physical quantities separated by the accumulated decoherence history of the matter sector.

5.3 Validation and falsification

Seventeen MCMC chains were run against the full Planck 2018 likelihood; all 17 converged to $R - 1 < 0.01$ with no discarded runs and no parameter adjustments. The best-fit improvement over Λ CDM is $\Delta\chi^2 = +0.54$ [Mahaffey, 2026b]. The decisive falsification test is Euclid DR1 (October 2026), which will measure μ_0 to $\sigma(\mu_0) \approx 0.04$ precision. IAM predicts $\mu_0 - 1 = -0.136$, a 3.4σ detection if correct, a definitive exclusion if absent.

6 Discussion

6.1 Time as the accumulated cost of classical existence

The framework presented here addresses a question that has resisted purely geometric treatment: why does time have a direction? The answer is not statistical. It is thermodynamic. Each quantum-to-classical transition pays an irreversible Landauer cost to the nearest available encoding surface. The accumulated total of these costs, written on the cosmic horizon, is $\mathcal{E}(a)$. Time has direction because actualization has direction. The ledger only grows.

This is consistent with the program of temporal naturalism [Smolin, 2013] in that it provides a physical mechanism for the irreversibility of time. It extends that program to a derivation: the arrow of time is not a fundamental postulate but a consequence of the thermodynamics of information production at every quantum-to-classical boundary.

It should be noted that Smolin’s temporal naturalism includes a stronger thesis: not only that time is real, but that the laws of physics themselves evolve in time [Smolin, 2013]. IAM does not make this claim. The thermodynamic laws governing $\mathcal{E}(a)$ — the first law, the second law, Landauer’s principle — are taken as fixed. The arrow of time emerges from the irreversibility of processes governed by those laws, not from the evolution of the laws themselves. Whether the deeper connection between IAM’s derived irreversibility and Smolin’s evolving-laws program constitutes a tension or a complementarity is an open question this paper does not resolve.

The laws governing $\mathcal{E}(a)$ are not timeless. They are written by the universe as it actualizes. $\mathcal{E}(a)$ is not a parameter of the theory imposed from outside — it is the theory’s own running record, carried forward by the universe as it becomes.

6.2 Dissipation-Driven Adaptation and the Virial Partition Closure Condition

England’s dissipation-driven adaptation [England, 2013, 2015] establishes a general thermodynamic principle: systems subject to a thermodynamic drive tend, over time, toward configurations that more efficiently absorb and dissipate energy from their environment. Systems that align with the thermodynamic flow endure; those that cannot dissipate sufficiently disperse. The probability of a driven system ending up in a given macroscopic configuration is proportional to the thermodynamic flux that occurs on the way to that configuration.

The virial partition closure condition derived in Section 4 is a specific gravitational instantiation of this principle. A collapsing dark matter halo is a driven system: gravitational potential energy is being converted irreversibly into kinetic degrees of freedom at the virialization boundary. The system endures as a galaxy if and only if it can route the

informational half of its virial partition to a local encoding surface — a black hole horizon. Where no such surface can form, the dissipation cannot complete, and the structure disperses. The virial partition closure condition is a necessary condition for endurance in the England sense, derived from first principles for gravitational $1/r$ systems.

This connection is not an analogy. The Landauer cost at the virialization boundary is the physical quantity England’s framework identifies as the thermodynamic flux required for stable self-organization. The IAM framework derives this cost explicitly for gravitational systems, locates the encoding surface that absorbs it, and produces the minimum velocity dispersion threshold $\sigma_{\text{crit}} \approx 4 \text{ km s}^{-1}$ as a falsifiable prediction. Whether the full England program — which addresses driven non-equilibrium systems far more generally, including biological self-organization — shares deeper mathematical structure with the IAM virial partition is an open question this paper does not address.

6.3 Thermal Time and Duration

Rovelli’s thermal time hypothesis [Connes and Rovelli, 1994, Rovelli, 1993] proposes that time emerges as a statistical effect from the thermodynamic state of a system: what we call the flow of time is the modular flow of a Gibbs state, and the direction of time is rooted in thermodynamic irreversibility rather than in any fundamental asymmetry of the laws of physics. The more recent program in Rovelli [Rovelli, 2019, 2022] asks specifically where the low entropy of the early universe was located and how thermodynamic traces constitute physical memory.

The duration variable $\mathcal{E}(a)$ derived in Section 3 is consistent with this program and extends it in a specific direction. Rather than time emerging from our incomplete knowledge of a statistical ensemble, duration in IAM emerges from the accumulated irreversible Landauer costs of individual quantum-to-classical transitions written permanently on the cosmic horizon. These are not statistical: each transaction is a specific physical event with a specific thermodynamic cost $k_B T_H \ln 2$. Duration accumulates not because we lack information about the microstate but because each decoherence event is physically irreversible at the level of the second law.

This extends the thermal time program in a specific direction: duration has a specific physical carrier ($\mathcal{E}(a)$), a specific beginning (the electroweak transition), a specific asymptotic limit (e), and specific cosmological observational consequences ($\mu(a) < 1$, $\sigma_8 = 0.800$, falsifiable by Euclid DR1). Whether this constitutes a derivation of a thermal time variable in Rovelli’s sense, or a complementary but distinct account of temporal irreversibility, is an open question the present paper does not resolve.

6.4 Cosmological Natural Selection and the IAM black hole role

Smolin’s cosmological natural selection [Smolin, 1992, 1997] proposes that black holes are reproductive units of baby universes, with constants selected by a natural-selection-like process across generations. IAM assigns black holes a different but complementary fundamental role: they are mandatory local encoding vaults, the thermodynamic structures that allow galaxies to exist at all by absorbing the informational half of their virial partitions.

Both frameworks place black holes at the center of cosmic structure. The question of whether these roles are independent, complementary, or aspects of a deeper unification is

an open problem that the IAM framework does not claim to resolve. It is noted here as a direction worth examining.

6.5 Open problems

The absolute normalization of $M_{\min}$ — the $\sim 100\times$ systematic offset between the predicted and observed satellite masses — requires the formal derivation of the local-to-global encoding rate criterion: the condition under which decoherence events are assigned to the local BH vault versus the global cosmic horizon. This is the same open normalization identified in the cusp-core problem [Mahaffey, 2026c], suggesting one unsolved problem rather than two. Its resolution will quantify the satellite count suppression from Mechanism B; the 13.6% growth suppression from $\mu < 1$ (Mechanism A) is derived and confirmed independently of this normalization.

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Data Availability

All MCMC chains, analysis scripts, and prediction figures are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> with canonical Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042). Repository timestamps confirm all predictions were established prior to any observational comparison.

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